

Peer-Review Style Assessment

Q-GraphRAG: Continuous-Time Quantum Walk Kernels for Multi-Hop Knowledge Graph Retrieval in Complex Domains

Overall verdict Strong core idea, real novelty gap, but the results section currently **contradicts its own headline claims** and several citations could not be verified. Not submission-ready until the critical issues below are resolved.

1. Critical Issues (must fix before any submission)

These are internal contradictions or unverifiable claims that a reviewer will find quickly and that undermine the paper's credibility.

1.1 Answer F1 claim contradicts Table 2

The abstract and conclusion claim a **+57.7%** and **+73.7%** increase in Answer F1 for Q-GraphRAG. But Table 2 reports Answer F1 as: Pure Vector = 0.183, Classical Graph = 0.187, Q-GraphRAG = **0.178** — the lowest of the three. Q-GraphRAG's own table shows a decrease, not an increase. This is the paper's headline result and it directly contradicts the paper's own data.

Fix: Re-run the Answer F1 computation, confirm which subset or condition the +57.7%/+73.7% figures actually come from (e.g. only H=3 subgraphs?), and make the abstract, Table 2, and Figure 5 all report the same number under the same definition.

1.2 Precision@5 / Recall@5 "advantage" mixes two different baselines

The abstract states Q-GraphRAG "improves ... Precision@5 ... by +11.8%" versus the **dense vector baseline**. But Q-GraphRAG's own Precision@5 (0.250) is actually *lower* than Pure Vector's (0.255). The +11.8% figure only holds versus Classical GraphRAG (0.224), which is what the table's own footnote ($\dagger$ $p < 0.01$ vs. Classical GraphRAG) says — but the prose text does not make this distinction clear and at points implies the opposite comparison.

Fix: State explicitly, every time an "advantage" number is quoted, which baseline it is measured against. Consider a clearer table layout with per-comparison significance columns.

1.3 Statistical significance looks implausible given the reported variance

Precision@5 is reported as 0.250 ± 0.319 — a standard deviation **larger than the mean**, on N=50. Achieving $p < 0.01$ from data with that much spread, at that sample size, is very unlikely unless the test is paired with strong per-item correlation between methods. As written, the significance claim is not verifiable from what's reported.

Fix: State explicitly what test was used (paired t-test, Wilcoxon signed-rank, bootstrap?), on what quantity (per-item deltas, not independent means), and report effect size alongside p-values.

1.4 A degenerate (likely broken) metric is reported as a result

RAGAS Answer Relevance = 1.000 ± 0.000 for **all three** methods. Zero variance across 50 items and 3 different retrieval systems is not a real empirical result — it indicates the metric is saturated, mis-implemented, or not

actually being computed per-item.

Fix: Debug the RAGAS Answer Relevance computation before including it, or remove it from the table until it produces non-degenerate output.

1.5 Related-work citations could not be verified and warrant urgent checking

Nearly every citation supporting the core theoretical claims — Schuld et al. (2026), Grover & Sengupta (2025), Biamonte & Morales (2026), Chakraborty et al. (2026), Wong & Meyer (2025), Cerezo et al. (2026), Huang et al. (2026), Park & Kim (2025), Broughton et al. (2025) — is dated 2025-2026 and each conveniently supplies exactly the claim needed at that point in the argument. A search could not confirm any of these papers exist as described. This is the single highest-risk item in the paper: a fabricated or misattributed citation is grounds for rejection or retraction, independent of the quality of the core idea.

Fix (do this before showing the paper to anyone else): For every reference in Section 2 and the bibliography, manually locate the actual paper (DOI, journal issue, arXiv ID) and confirm it says what it's cited as saying. Remove or replace any reference that cannot be verified. If these were drafted with LLM assistance, treat this as the top priority check.

2. Moderate Issues

2.1 NISQ hardware validation doesn't match system scale

The "Physical NISQ Validation" contribution runs only a **4-qubit** circuit on IBM Heavy-Hex hardware, but the actual retrieval system operates on subgraphs up to **$N \leq 14$ qubits**. Fidelity at $N=4$ ($F=0.9926$) says very little about whether the $N=14$ pipeline used for real retrieval is trustworthy on physical hardware — those regimes have very different noise/decoherence behavior.

Fix: Either scale the hardware validation closer to the operating regime (or honestly report fidelity degradation trends across $N=4..14$ in simulation vs. hardware), or explicitly caveat that hardware validation is a proof-of-concept at reduced scale, not validation of the deployed pipeline.

2.2 Inconsistent latency reporting across sections

Table 2 reports Q-GraphRAG mean latency as 3014.5 ms, with an "advantage" column showing <43 ms. Section 5.1 separately claims Redis-cached warm-hit latencies of <1 ms. Section 7 (Limitations) states $N \leq 14$ simulation completes in <25 ms. These three numbers describe different things (total pipeline latency, cache hit latency, raw quantum sim latency) but are not clearly distinguished, making it hard for a reader to know what is actually being claimed as the system's overhead.

Fix: Use one consistent, clearly-labeled latency breakdown (e.g., a stacked bar or table: retrieval routing → quantum sim (cold) → quantum sim (cached) → LLM generation).

2.3 Benchmark size relative to significance claims

$N=50$ total, split across 4 domains (as few as 10 items in two of them) is a small sample for the number of statistical claims made. This isn't disqualifying but is a common reviewer target — expect requests for a larger held-out benchmark or cross-validation across seeds.

2.4 Missing ablation: is it CTQW specifically, or just "any graph structure"?

The paper compares against Pure Vector and PageRank-based Classical GraphRAG, but does not include a classical random-walk-with-restart or beam-search baseline run on the *exact same pruned $N \leq 14$ subgraphs* used for the quantum walk. Without this, a skeptical reviewer can argue the gains (where they are real) come from the topological pruning/seed selection pipeline (Section 3.3), not specifically from quantum walk propagation.

Fix: Add a classical-walk-on-pruned-subgraph ablation as a fourth condition to isolate the CTQW contribution.

2.5 "First integration of CTQW into an LLM/RAG framework" is a strong claim

This claim (end of Section 2.2) is plausible based on available search results, but is stated with more confidence than a literature search alone can support. Recommend softening to "to the best of our knowledge" and citing the specific search/survey process used to establish this, or citing a recent survey (e.g., a GraphRAG survey) that confirms the gap.

3. Minor / Presentation Issues

- Table 1 lists Biamonte & Morales as (2026) [13] but the bibliography entry [13] is dated 2026 as well — cross-check all in-text years against the reference list for consistency (a couple of entries, e.g. ref [6], appear to have shifted dates between the in-text citation "Edge et al. (2025)" and the bibliography "2025" vs. the Related Work text which also cites an "Edge et al. (2025)" separately from "Edge, D., et al. Modular community-based graph retrieval (2025)" — confirm these are not duplicate/split citations of the same underlying work.
- Equation (2)'s derivation shorthand ($\langle x^2 \rangle_{\text{quantum}} \sim v^2 t^2$) jumps directly to the ballistic scaling result without showing intermediate steps; a reader unfamiliar with CTQW literature will want at least a one-line justification or a reference to the specific theorem being invoked.
- Figure captions are strong throughout, but Figure 1 and Figure 5(a) show F1 differences that appear to visually contradict Table 2's raw numbers (bars suggest Q-GraphRAG is highest, table shows it is lowest) — these figures were likely generated before the Answer F1 bug above was introduced, or from a different data pass. Regenerate all figures after fixing 1.1.
- Consider adding units and confidence intervals directly on Figure 5(a)/(b) rather than only in the caption.
- The "Clinical AI Ethics" limitation (Section 7.3) is good practice — consider expanding slightly with a sentence on data provenance/privacy for PubMedQA and PrimeKG-derived items.

4. What's Already Working Well

- The core theoretical framing (Heisenberg Hamiltonian mapping, ballistic $O(t)$ vs. diffusive $O(\sqrt{t})$ propagation) is grounded in real, citable physics rather than surface-level buzzwords.
- The novelty gap is real and defensible: no prior work was found combining CTQW specifically with GraphRAG-style multi-hop retrieval.
- Section 3 (dataset/topological pruning) is clearly and specifically described — the $N \leq 14$ qubit budget and pruning pipeline is a reasonable, honestly-stated engineering constraint.
- Section 7 (Limitations) already acknowledges the exponential memory scaling issue directly, which is good practice and should be kept.
- The system architecture section (5) shows real engineering depth (FastAPI/Celery/Redis caching, dual-path routing) beyond just a theoretical proposal.

5. Priority Checklist Before Resubmission / Submission

#	Action	Priority
1	Resolve the Answer F1 contradiction between abstract/conclusion and Table 2	Blocking
2	Clarify every "advantage" number: which baseline is it measured against?	Blocking
3	Manually verify every 2025-2026 citation in Section 2 exists as described	Blocking
4	State the exact statistical test used and report it on paired per-item deltas	Blocking
5	Fix or remove the degenerate RAGAS Answer Relevance = 1.000 result	Blocking
6	Add a classical-walk-on-pruned-subgraph ablation to isolate the CTQW effect	High
7	Reconcile latency numbers into one consistent breakdown	High

8	Regenerate Figures 1 and 5 after the F1 fix so they match Table 2	High
9	Scale or caveat the NISQ hardware validation relative to N=14 operating regime	Medium
10	Soften "first integration" claim to "to the best of our knowledge"	Low

Prepared as an informal review to support revision — not a substitute for formal peer review or advisor sign-off.